

Multi-mode Microring Spectrometer without Photodetector

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Abstract We demonstrate a multi-mode microring resonator spectrometer assisted with two-photon absorption, achieving a 40 pm resolution across 50 nm bandwidth and record-high channel-number-to-footprint ratio of $2.8 \times 10^6 \text{ mm}^{-2}$. Without any external photodetector, this fully integrated spectrometer demonstrates excellent capability in on-chip optical sensing applications. ©2026 The Author(s)

Introduction

Miniaturized spectrometers have merged to be critically potential in in-situ and portable applications [1]. In recent years, the reconstructive spectrometer (RS) has become a new miniaturization paradigm through elaborate encoding strategy [2]. Via various sampling channels with distinct spectral response, the RS employs miniaturized encoding hardware to map the entire incident spectrum into highly encoded measurable signals, and reconstructing spectral profiles by combining the separated decoding devices with computational algorithms. This approach leads to the emergence of the high-performance RSs within ultra-compact footprint [3-5]. Despite these RSs have achieved the miniaturization in encoding devices, they still require external photodetectors to measure the encoded signals. The modern foundry processes provide the possibility in the monolithic integration of photodetectors [6],

however, requiring larger footprints and costs. The development of low-cost, miniaturized spectrometers remains a key goal, as their full integration is essential for advancing high-density photonic integrated circuits.

In our previous work, a two-photon absorption (TPA) spectrometer without an external photodetector was realized, featuring a bandwidth of 10 nm and a resolution of 0.4 nm [7]. However, the limitations in bandwidth and resolution remain a significant challenge that restricts practical applications.

In this paper, we propose a novel RS that utilizes a multi-mode microring resonator (MRR) assisted by TPA effect, enabling spectrometer function without additional photodetectors while occupying an ultra-small footprint of $21 \times 21 \mu\text{m}^2$. We experimentally demonstrate a 50 nm bandwidth and a 40 pm resolution utilizing multi-mode resonance, breaking both the free spectral range

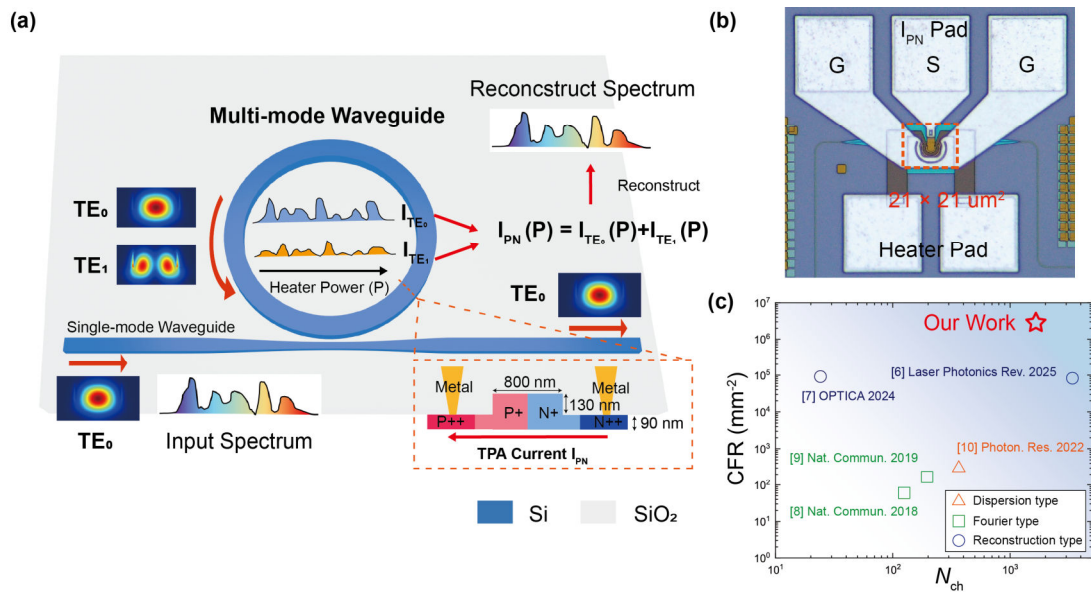


Fig. 1: (a) Schematic of the multi-mode microring resonator (MRR) spectrometer based on the two-photon absorption (TPA) current I_{PN} . (b) Microscope image of the multi-mode MRR spectrometer. (c) Comparison of typical fully on-chip spectrometers regarding channel-number-to-footprint ratio (CFR) versus channel-number N_{ch} (N_{ch} = Bandwidth / resolution).

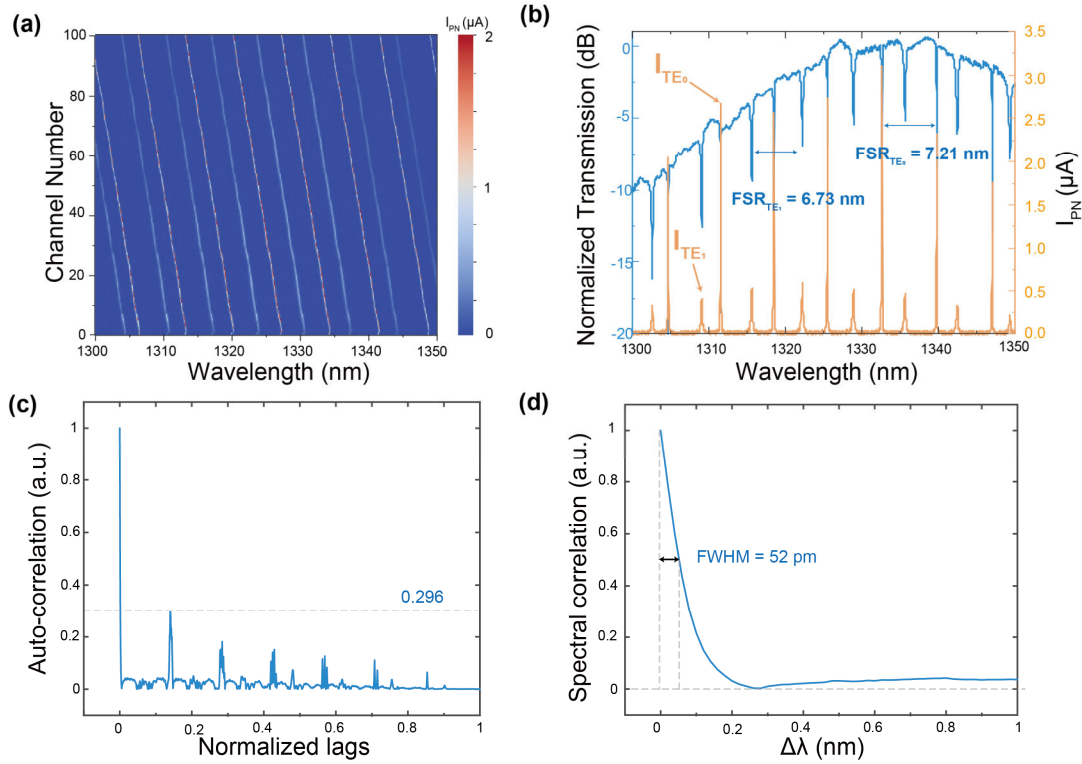


Fig. 2: (a) Response matrix obtained by linearly sweeping the heater power from 0 to approximately 100 mW and the wavelength from 1300 nm to 1350 nm across 100 heating channels. (b) Normalized transmission and TPA current I_{PN} of the multi-mode MRR spectrometer at channel 72. (c) Normalized autocorrelation function, where the periodicity level is indicated by the grey dashed line. (d) Normalized spectral correlation function, where the estimated resolution is marked by the grey dashed line.

(FSR) limit and the quality factor (Q) limit inherent in a single MRR. As shown in Fig. 1c, we demonstrate a record-high channel-number-to-footprint ratio (CFR) of $2.8 \times 10^6 \text{ mm}^{-2}$, which is more than 20 times higher than that of existing fully integrated spectrometers—whether they employ monolithically integrated photodetectors or rely on the TPA effect [6-10]. Spectrometers with external photodetectors are excluded here, as the unaccounted photodetector area would unfairly increase the CFR, and the use of external photodetectors also incurs substantial hidden costs, such as compromising system compactness.

Multi-mode MRM spectrometer design and experiment results

The multi-mode MRM spectrometer was fabricated by a 220 nm silicon-on-insulator multi-wafer project (MPW), which provided by advanced micro foundry (AMF). Fig. 1a shows the schematic of the multimode MRM spectrometer. The multi-mode MRR, featuring an 800 nm width and a 9 μm radius, is coupled to a 400 nm wide bus waveguide via a taper coupling region, which ensures strong coupling between the fundamental mode in the bus waveguide and the TE₀ and TE₁ modes in the MRR. The PN junction in MRR is a two-level lateral doping: N^+ -type doping with the concentration of $3 \times 10^{18} \text{ cm}^{-3}$, P^+ -type doping of $3 \times 10^{18} \text{ cm}^{-3}$, N^{++} -type doping of $1 \times 10^{20} \text{ cm}^{-3}$,

and P^{++} -type doping of $1 \times 10^{20} \text{ cm}^{-3}$. Fig. 1b shows the microscope image of the spectrometer with the five pads configuration. The TPA current I_{PN} is measured under a 2 V reverse bias voltage between the two ground pads and one signal pad. The heater which is on the top of the MRR is utilized to measure the I_{PN} of different 100 heating channels ranging from 0 mW to 100 mW.

Here, we use a tunable laser with a 20 pm tuning step and two source meters (Keithley 2400) to acquire the response matrix shown in Fig. 2a: one for heater power control and the other for current measurement. The normalized transmission and I_{PN} of heating channel 72 are shown in Fig. 2b. The natural two modes differences in the FSR and I_{PN} , both in spectrum and in magnitude, arising from multi-mode resonance, improve the bandwidth and resolution of the RS spectrometer. The full-width at half maximum (FWHM) of the TE₀ and TE₁ resonance are 90pm and 230pm respectively. Fig. 2c shows the calculated average auto-correlation. With an effectively suppressed periodicity measured at 0.296, the response matrix exhibits sufficient diversity induced by multi-mode resonance. The estimated resolution shown in Fig. 2d is shown as 52 pm, which is mainly caused by the square relationship between resonance power and I_{PN} .

In Fig.3a, a series of single-peak signals is generated from a tunable laser source across the

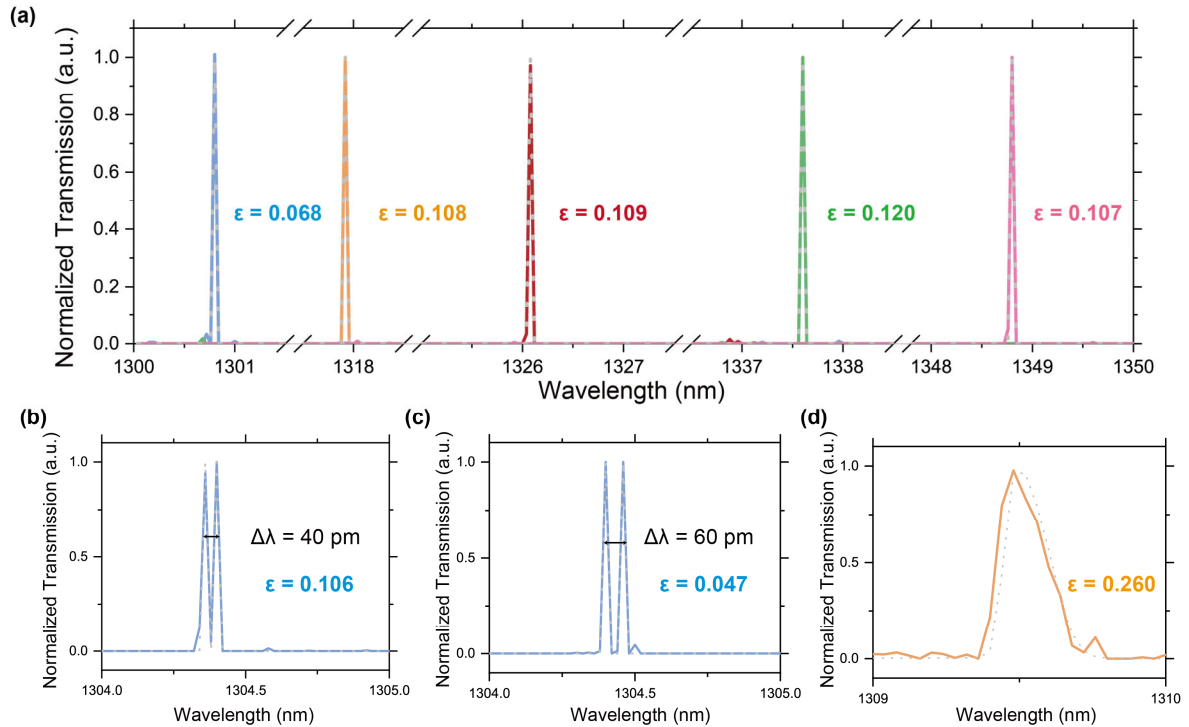


Fig. 3: (a) The reconstructed spectrum of single laser peaks across a 50 nm bandwidth, with the dashed grey lines marking the narrow laser peak. (b, c) Reconstructed dual-laser peak with different wavelength intervals of 40 pm and 60 pm. (c) The reconstructed spectrum of a continue signal.

1300–1350 nm wavelength for the single-peak reconstruction. The relative error ϵ , ranging from 0.068 to 0.120, demonstrates good spectrum reconstruction over the 50 nm bandwidth. To determine the resolution of the multi-mode MRR spectrometer, we employ dual-laser peaks with wavelength spacings of 40 pm and 60 pm, as shown in Fig. 3b and Fig. 3c, respectively. The low relative error ϵ of 0.106 demonstrates a 40 pm resolution of the RS spectrometer. In Fig. 3d, a continuous spectrum reconstruction is shown, with a relative error ϵ of 0.26. The resolution of the spectrometer can be further improved by reducing the wavelength step from 20 pm to 1 pm and by increasing the number of heating power channels, though such improvements are primarily limited by the trade-off between total acquisition time and reconstruction resolution. Replacing the current grating coupler, which exhibits a 7 dB per facet loss, with an edge coupler would enable further improvement of the spectrometer bandwidth.

Conclusions

We demonstrate a fully integrated reconstructive spectrometer with a 40 pm resolution and a 50 nm bandwidth in an ultra-small footprint of $21 \times 21 \mu\text{m}^2$ without any extra photodiodes, realizing a record-high CFR of $2.8 \times 10^6 \text{ mm}^{-2}$. The multi-mode MRR spectrometer, which exhibits both TE0 and TE1 resonances, overcomes the FSR-induced bandwidth limitation that constrains single-mode MRR spectrometer. This fully

integrated spectrometer enables on-chip optical sensing, thereby boosting the development of lab-on-a-chip systems and enabling low-cost portable sensing modules.

Acknowledgements

This work is supported by the National Key Research and Development Program of China under contract nos. 2023YFB2804702 (granted to X.G.), in part by the Natural Science Foundation of China under contract nos. 62175151 (granted to X.G.) and 62341508 (granted to Y. S.), and in part by Shanghai Science and Technology Innovation Action Plan under contract nos. 25LN3201000 (granted to X.G.) and nos. 25JD1405500 (granted to X.G.) and Shanghai Municipal Science and Technology Major Project under contract nos. BH030071 (granted to Y. S.).

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